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A HETEROGENEOUS AGENT NEW KEYNESIAN MODEL WITH BIASED LABOR MARKET EXPECTATIONS: IMPLICATIONS AND CONCLUSIONS

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A Heterogeneous Agent New Keynesian Model with Biased Labor Market Expectations: Implications and Conclusions

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Abstract

I introduce biased labor market expectations in a heterogeneous-agent New Keynesian model and simulate a positive technology and a contractionary monetary policy shock for two cases. In the first case, I calibrate the entire model for the U.S., where the population exhibits correct labor market expectations when employed but displays biased, optimistic expectations when unemployed. In the second case, I take Germany as a calibration target for my model. Unlike in the U.S., Germans tend to be more pessimistic about losing their jobs when they are employed. In contrast to technology shocks, a monetary policy shocks dampens the response of output, unemployment, and real interest rate, and amplifies the response of the nominal interest rate, job finding rate, and inflation rate through biased labor market expectations in the first case. The two cases exhibit reversed effects when introducing aggregate shocks.

Keywords: Aggregate risk, Heterogeneous Households, Behavioral Macroeconomics

JEL Classification: E21, E37, E50, E71

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1 Introduction

The representative-agent commonly used in most Dynamic stochastic general equilibrium models seems to struggle in capturing the heterogeneity within the economy. Recent developments following Mankiw [2000] led to the introduction of more heterogeneity in the form of two types of agents. However, this framework does not introduce enough flexibility to capture the different decisions made by various types of agents. This led to the creation of a novel model known as the heterogeneous-agent New Keynesian (HANK) model, which emerged by combining two fundamental pillars of macroeconomic theory: the New Keynesian approach to study business cycles and stabilization policies, and the incomplete-market approach to study the distribution of income and wealth. The HANK literature has contributed to a deeper comprehension of how monetary policy actions impact the economy. In recent years, efforts to enhance HANK models' accuracy in replicating real-world dynamics have grown by incorporating behavioral elements. These papers utilize established behavioral biases from the economics literature, primarily derived from experimental research. In recent studies, there has been comprehensive research in the expectations that individual households hold about labor market outcomes. These expectations cover a range of factors, including workers' predictions about potential job loss, wage increases, and finding new jobs.[1] In contrast to the rational expectations theory, these studies' findings present convincing proof of significant and enduring biases in individual labor market expectations. For example, Mueller et al. [2021] use data from the Survey of Consumer Expectations to show that job seekers in the United States tend to overestimate their chances of find-

[1] Mueller and Spinnewijn [2023] conducted an extensive survey that offers valuable insights into this area.

ing employment, compared to the actual statistical probability. Likewise, Spinnewijn [2015] reveals similar outcomes for job seekers in the United Kingdom, and Balleer et al. [2023] identify comparable trends among job seekers in Germany. Additionally, Balleer et al. [2021] show that unemployed individuals in the United States consistently overestimate the chances of becoming employed. On the other hand, Balleer et al. [2023] find a contrasting pattern among German workers, who consistently hold a pessimistic view about job stability. Considering this context, "bias in expectations" refers to the disparity between the subjective transition probability perceived by workers and the corresponding statistical probabilities. An optimistic bias in job finding expectations arises when workers overestimate the probability of moving from unemployment to employment. Conversely, a pessimistic bias in job separation expectations emerges when workers overestimate the likelihood of shifting from employment to unemployment.

In this paper, I combine both a heterogeneous-agent framework based on Ravn and Sterk [2021] with an endogenous Markov chain for idiosyncratic risk based on Blanchard and Galí [2010], which introduces countercyclical income risk. Households are subject to biases in their labor market expectations. Motivated by Balleer et al. [2021], based on U.S. data, households are too optimistic to find a job when unemployed, which has consequences for their savings decision and thus the steady-state wealth distribution. In this case, due to weakened precautionary savings motive, more households are at the borrowing constraint which makes monetary policy less effective. However, a technology shock on the other hand, has larger impact on aggregate output since the economy is comprised of more high marginal propensity to consume (MPC) households. This polarity comes through the different channels of how the shocks enter the economy.

Monetary policy becomes less effective because overall wealth holdings are lower, and thus unexpected changes in interest rates affect the value of assets in absolute terms less. The technology shock, on the other hand, enters the economy through household's income. Consequently, the impulse response functions for the economy with biased labor market expectations substantially differs from the economy with rational expectations for a monetary policy as well as a technology shock. Furthermore, I explore a relatively recent data set by the European Central Bank (ECB) which covers six EU countries and asks its participants questions regarding job market expectations. Based on this survey, it is possible to study labor market expectations for additional countries that have not been considered in the literature so far. I find for all six countries evidence that unemployed workers are too optimistic to find a job and employed workers are too pessimistic to lose their job in the aggregate. Following this, I simulate an economy with labor market expectations calibrated based on Germany. The introduction of pessimism when employed strengthens the precautionary savings motive which results in fewer mass of agent at the borrowing constraint and mass of high MPC agents. Compared to the calibration for the U.S., I obtain reversed effects when introducing aggregate shocks.

Layout The remained of this paper is organised as follows. Section 2 provides a broad literature review which outlines the current developments of HANK models with non-rational expectations. Section 3 explores ECB's Consumer Expectation Survey. Section 4 introduces the model. Section 5 discusses the model's calibration and its quantitative results, and Section 6 concludes.

2 Literature Review

In recent years, there has been an increasing push to incorporate behavioral elements into HANK models, aiming to improve their accuracy in replicating real-world dynamics. This effort began with the work of Farhi and Werning [2019], who incorporate agent heterogeneity with incomplete markets, uninsurable idiosyncratic risk, and occasionally binding borrowing constraints and level-k thinking into a benchmark HANK model. Relative to the representative-agent New Keynesian model, they show that the interaction of level-k thinking, i.e. bounded rationality, and incomplete markets makes monetary policy less effective and may offer a possible resolution of the "forward guidance puzzle". Auclert et al. [2020] estimate a HANK model including sticky expectations on the household side which corresponds to the existing microeconomic evidence on MPCs and macroeconomic support on the impulse response to a monetary policy shock. Since high MPCs amplify the investment response in the data, they highlight the central role for investment in the transmission mechanism of monetary policy, as high MPCs amplify the investment response in the data. Additionally, this force gives rise to a cyclical pattern in which changes in consumption closely follow fluctuations in investment due to shocks. This leads their model to suggest that these shocks play a key role in driving business cycles. Laibson et al. [2021] examine the impact of monetary and fiscal policies within a HANK model that incorporates present-biased time preferences and naive beliefs on the household side. Households can invest in a liquid asset and illiquid home equity, where the illiquid asset serves as a collateral for borrowing. Since present bias considerably increases households' MPC, the present bias increases the impact of fiscal policy and also amplifies the effect of monetary policy, but reduces

the velocity of monetary transmission. Pfäuti et al. [2023] demonstrate that households possessing lower cognitive skills consistently exhibit excessive confidence in their own abilities, too positive about their future finances, and have a higher tendency to be hand-to-mouth. In their paper, they incorporate permanent heterogeneity in households' cognitive skills and beliefs about their skills in a HANK model to match these findings. Their model matches U.S. average marginal propensities to consume and the average wealth. Pfäuti and Seyrich [2022] examine the impact of cognitive discounting and heterogeneity among households on the propagation of monetary policy shocks within the economy. Under cognitive discounting, households' expectations demonstrate a tendency to react less to information about the aggregate economy, which is in line with real-world observations of household expectations. Their model accommodates empirical findings regarding the propagation of monetary policy, e.g. monetary policy influences consumption largely through indirect effects, households are unequally exposed to aggregate fluctuations, and forward guidance is less powerful.

Contribution to the literature I contribute to a growing literature that introduces non-rational expectations into a HANK framework to examine how deviations from rational expectations amplifies and dampens certain aggregate variables as a response to monetary policy and technology shocks. Specifically, I introduce biased labor market expectations, another prominent bias in the labor economics literature, into a HANK framework. Balleer et al. [2021] estimate the implications of biased labor market expectations in a Bewley-Huggett-Aiyagari type model, to analyse their effects on the steady-state wealth distribution. I am the first to study the implications of biased labor market expectations on the impulse response functions of monetary policy and

technology shocks. The paper closest to mine is Pfäuti et al. [2023], who, by including overconfidence, can better match MPCs and wealth holdings in the data and study the transmission effects of monetary policy when non-rational expectations on the household side are present.

3 Data

In this section, I explore ECB's Consumer Expectation Panel (CES) - a quarterly survey that represents households in Belgium, Germany, Spain, France, Italy, and the Netherlands - for my further analysis of biases over labor market outcomes. The data is an unbalanced panel that covers the years from 2020 to 2023.[2] It covers questions on expectations on household level to individual level and expectations of the state of the economy. The total number of observations is 162,853 and I have 32,918 distinct households from six countries. Within Europe, the CES is unique since it is possible to compare the difference in household's expectations formation between countries. The individual quarterly CES survey includes multiple questions about expected individual labor market transitions.

Expectations about Labor market transitions Starting from the first wave in 2020, respondents who are employed at the time of the interview are asked "What do you think is the percent chance that you will lose your current job during the next 3 months?" upon which they should assess the probability on a 0 to 100 scale. Using the corresponding variable for the answer on job loss provides us with a direct measure of an individual's expected job separation probability. Similarly, since 2020

[2]The data set is available at: https://www.ecb.europa.eu/stats/ecb_surveys/consumer_exp_survey/html/data_methodological.en.html

respondents who are not working at the time of the interview are asked "Please think about the type of job that may be available to you. What do you think is the percent chance that, within the next 3 month, you will find a job that you will accept?" upon which they should assess the probability of taking up a job on a scale from 0 to 100 percent. Using the corresponding variable for the answer on taking up a job provides us with a direct measure of an individual's expected job finding probability. Employed workers, on average, state a 10.8% probability to separate from their job within the next quarter. Job separation expectations are very dispersed, include the full range from 0 to 100 percent probability. For all countries the distribution of expected job market probabilities is right-skewed. Unemployed workers, on average, state a 26.7% probability to find a job within the next quarter. Also job finding expectations are very dispersed, including the full range of 0 to 100 percent probabilities. They are, however, more uniformly distributed than job separation expectations.

Average expected job finding rate for the next 3 month for individuals who are unemployment and "actively looking for a job", "not actively job looking for a job", and both aggregated conditional on country, gender and level of education are summarised in Table 8, Table 9 and Table 10 respectively. Average expected job separation rate for the next 3 month conditional on country, gender and level of education is summarised in Table 11.[3]

In my first regression, summary in Table 12, I regress expected job separation rate for employed individuals on disposable income quintile, level of education, gender, country, presence of a partner in a pooled linear panel regression.[4] Individuals with a higher

[3] 1=Male, 2=Female; 1=low education,2=medium education,3=high education; BE=Belgium, DE=Germany, ES=Spain, FR=France, IT=Italy, NL=Netherlands.

[4] quintile $\in \{1, ...10\}$, level of education $\in \{1, 2, 3\}$, gender $\in \{1, 2\}$, country $\in$

education significantly expect a higher job separation rate. Furthermore, participants from Spain, Italy, and Netherlands are significantly expecting a higher job separation rate than participants in Belgium on average. Only participants from France expect a lower job separation rate than participants from Belgium on average. The estimator for Germany is insignificant. Moreover, on average, the fifth quintile in the survey significantly expects a higher job separation rate than the first quintile.

In my second regression, summary in Table 13, I regress expected job finding rate for searching unemployed individuals on disposable income quintile, level of education, gender, country, presence of a partner in a pooled linear panel regression. Participants in the fifth quintile significantly expect a higher job finding rate than participants in the first quintile on average. Women in the sample significantly expect a lower job finding rate, and individuals with a partner significantly expect a lower job finding rate than their single counterpart on average.

Predicted labor market transitions Due to the panel structure of the data, I can identify actual job separations and job finding events within a period of 3 month following their interview. Individuals who are either "working full-time", "working part-time", "temporarily laid-off" or "on extended leave unconditional version" are considered as employed. Individuals who are either "unemployed and actively looking for a job" or "unemployed, interested in having a job but not actively looking for a job" are considered as unemployed. Lastly, individuals who are unable to work because of disability or other medical reasons, in retirement or early retirement, studying, at school, or in training, looking after children or other persons, doing housework or other fall in

$\{BE, FR, GER, IT, NL, ES\}$, presence of a partner $\in \{1, 0\}$. Same explanatory variables for the second regression.

the category "other".

Bias in labor market expectations I estimate actual job separation and job finding rates by sample means of realized labor market transitions conditional on the country. Based on that, I obtain predicted probabilities of losing a job or finding a job for each country in the survey. In order to compare them to my measured expectations, bias in labor market expectations is then defined as the difference between individual expected job separation and job finding probabilities and their statistical (predicted) counterpart. Table 1 documents summary statistics for expected and predicted job separation and job finding probabilities as well as the resulting biases. The resulting biases show that optimism and pessimism coexist in the sample. For the aggregate job separation measure, employed workers are predicted to separate from their job within the next three month with an average probability of 1.4% while they predict a probability of 10.8%. Regarding job finding out of unemployment, workers are predicted to find a job within next quarter with an average probability of 14.1%, while they expect to do so with a probability of 26.7%. Hence, unemployed workers, on average, are too optimistic regarding job finding, as they overestimate the chance of finding a job in this time interval.

	Actual Average Job separation rate	Actual Average Job finding rate	Bias in separation rate	Bias in finding rate
Belgium	0.94 %	12.97 %	8.02 pp	11.33 pp
France	0.91 %	10.90 %	8.09 pp	17.1 pp
Germany	0.84 %	15.38 %	7.18 pp	10.92 pp
Italy	1.74 %	13.34 %	9.46 pp	9.06 pp
Netherlands	0.63 %	15.46 %	7.23 pp	5.24 pp
Spain	2.05 %	16.87 %	14.05 pp	15.03 pp

Table 1: Statistical transition probabilities and biases in CES

4 Model

In the model, time is discrete and indexed by $t = 0, 1, 2, ...$, and agents face an uninsurable idiosyncratic shock which impacts their labor income and can be interpreted as being employed or unemployed. In each period, they might change from a status to another according to a transition matrix. The economy is composed of one unique profit-maximizing firm which produces a final good, and of a continuum of intermediate firms of size 1. There is one asset in the economy, which yields an endogenous real interest rate, and each agent saves and consumes. Households take their decisions according to a time-separable intertemporal utility function. They are constrained in their decisions both by their budget and by their borrowing constraint. In this model, there is a government as well as a common monetary authority, a central bank, which chooses the nominal interest in each period based on current inflation. In equilibrium, the bond and goods market clear. The economy is subject to two aggregate shocks: a technology and a monetary policy shock. The technology shock is denoted as $\epsilon_{A,t}$, and total factor productivity (TFP) by A_t, and I assume that it follows a $AR(1)$ process. Regarding the monetary policy shock, I denote the monetary policy shock as $\epsilon_{\xi,t}$ and assume that ξ_t follows a $AR(1)$ process.

Households There is a unit mass of households that are subject to idiosyncratic risk, incomplete market, and borrowing constraint. Each agent can save and consume, and labor supply is exogenous. The household maximizes

$$E_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma} \tag{1}$$

subject to the budget constraint in real terms

$$a_t + c_t = R_t a_{t-1} + earnings(e_t) \tag{2}$$

if the agent is employed, $e_t = 1$, the earnings are $(1 - \tau_t)(w_t + d_t)$, otherwise, $e_t = 0$, she receives time invariant unemployment benefits b. The income tax is denoted by τ_t, wage by w_t, and dividends from firms by d_t. In addition, $a_t \geq 0$ is the borrowing constraint.

The transition matrix across employment states with biased labor market expectations is given by

$$\begin{bmatrix} 1 - (M_t + \Delta_2) & (\delta + \Delta_1) \\ (M_t + \Delta_2) & 1 - (\delta + \Delta_1) \end{bmatrix}$$

where, column-wise, the first state is unemployment and the second is employment. M_t is the actual probability of finding a job, whereas $M_t + \Delta_2$ is the perceived probability of finding a job, and δ is the actual probability of separating from a job, whereas $\delta + \Delta_1$ is the perceived probability of separating from a job. In its recursive form, the household problem can be formulated as

$$\begin{aligned} V(e, a) = \max_{a'} \quad & \frac{(Ra + earnings(e) - a')^{1-\gamma}}{1 - \gamma} + \beta E[V(e', a')|e] \\ \text{s.t.} \quad & a' \geq 0 \end{aligned} \tag{3}$$

Figure 1 shows the different consumption functions for the agents for the different calibrations, which I will discuss in Section 5.1.

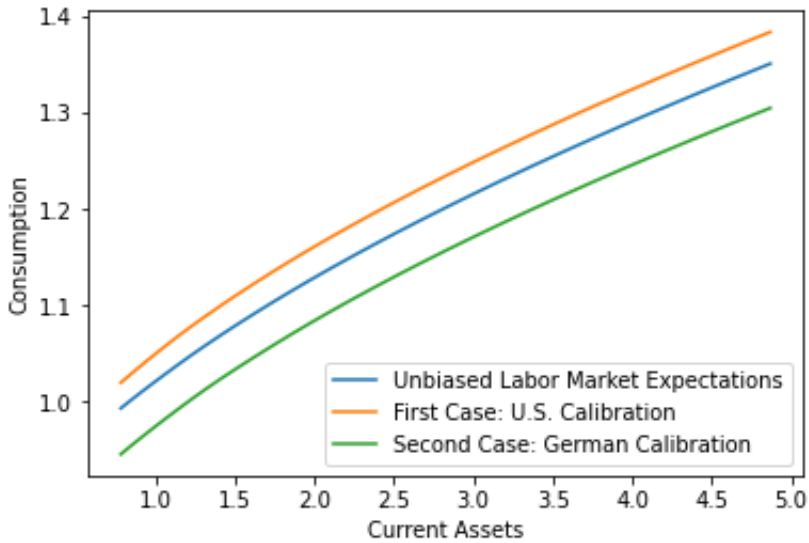

Figure 1: Consumption function for different calibrations for agents in the employment state

Firms Domestic production Y_t is aggregated into a domestic good by a unique profit-maximizing representative firm using a Dixit-Stiglitz aggregator, with an elasticity of substitution denoted by θ. The final good firm combines intermediate goods $y_t(J)$ from different sectors of the economy indexed by $J \in [0, 1]$. Therefore, the production Y_t can be expressed using a CES aggregator of domestic individual productions such that:

$$Y_t = [\int_0^1 y_t(J)^{\frac{\theta-1}{\theta}} dJ]^{\frac{\theta}{\theta-1}} \tag{4}$$

The final good firm solves the Dixit-Stiglitz cost-minimization problem resulting in a demand for variety J of

$$y_{J,t} = (\frac{p_{j,t}}{P_t})^{-\theta} Y_t \tag{5}$$

and the price index P_t is given by

$$P_t = [\int_0^1 p_t(J)^{1-\theta} dJ]^{\frac{1}{1-\theta}} \tag{6}$$

The intermediate goods firm, which is subject to nominal rigidities à la Calvo, updates its price each period with probability $1 - \Theta$ to maximize its profits

$$E_t \sum_{s=t}^{\infty} \Theta^{s-t} R_{t,s}^{-1} [\frac{p_t*}{P_s} y_{J,s} - w_s n_{J,s} - \psi M_s h_{J,s}]. \tag{7}$$

subject to

$$y_{J,s} = (\frac{p_t*}{P_s})^{-\theta} Y_s \tag{8}$$

$$y_{J,s} = A_s n_{J,s} \tag{9}$$

$$n_{J,s} = (1 - \delta) n_{J,s-1} + h_{J,s} \tag{10}$$

where (8) is the demand from the final good producer, (9) denotes the production function, and (10) is the law of motion for firm specific employment.

The first order conditions of the Lagrangian from the firm's optimization problem out-

lined above with respect to p_t^*, $y_{J,s}$, $n_{J,s}$, and $h_{j,s}$ are

$$E_t \sum_{s=t}^{\infty} \Theta^{s-t} R_{t,s}^{-1} [\frac{y_{J,s}}{P_s} - \lambda_{1,s} \theta (\frac{p_t*}{P_s})^{-\theta-1} \frac{Y_s}{P_s}] = 0 \tag{11}$$

$$\frac{p_t*}{P_s} - \lambda_{1,s} - \lambda_{2,s} = 0 \tag{12}$$

$$-w_s + \lambda_{2,s} A_s - \lambda_{3,s} + (1-\delta) E_s \lambda_{3,s+1} = 0 \tag{13}$$

$$\lambda_{3,s+1} - \psi M_s = 0 \tag{14}$$

Combining the first order conditions yields the optimal pricing decision:

$$\frac{p_{J,t}*}{P_t} = \frac{\Omega_t}{\upsilon_t} \tag{15}$$

where

$$\Omega_t = \frac{\theta}{\theta - 1} \frac{w_t + \psi M_t - (1-\delta) E_t[\psi M_{t+1}]}{A_t} Y_t + \Theta E_t [R_{t+1}^{-1} \pi_{t+1}^{\theta} \Omega_{t+1}] \tag{16}$$

and

$$\upsilon_t = Y_t + \Theta E_t [R_{t+1}^{-1} \pi_{t+1}^{\theta-1} \upsilon_{t+1}] \tag{17}$$

Recall, that a fraction, $1 - \Theta$, of prices are updated each period.

$$P_t^{1-\theta} = \Theta P_{t-1}^{1-\theta} + (1 - \Theta)(p_t^*)^{1-\theta} \tag{18}$$

Thus, rearranging further yields the equation for the New Keynesian Philips curve.

$$\pi_t = \Theta^{\frac{1}{1-\theta}}[1 - (1 - \Theta)(\frac{p_t^*}{P_t})^{1-\theta}]^{\frac{-1}{1-\theta}} \tag{19}$$

Monetary policy and Government There are three aspects of policy in this economy. First, there is social insurance: employed workers pay a proportional labor income tax with rate τ_t and unemployed workers receive an unemployment benefit in amount b. Second, the government has a constant real stock of debt B outstanding, which it rolls over in perpetuity. The interest on the debt is financed by the labor income tax τ_t. The government budget balance has to hold for all t:

$$\tau_t(w_t + d_t)(1 - u_t) + B = R_t B + bu_t \tag{20}$$

Finally, there is a monetary policy in the form of a rule for the nominal interest rate, where ξ_t is a monetary policy shock that follows a $AR(1)$ process in logs.

The central bank is characterized by the Taylor rule

$$1 + i_t = \bar{R}\pi_t^w \xi_t \tag{21}$$

The Fisher equation is

$$R_t = \frac{1 + i_{t-1}}{\pi_t} \tag{22}$$

Labor market I will assume the wage is an increasing function of labor market tightness M_t,

$$w_t = \bar{w}(\frac{M_t}{\bar{M}})^{\zeta} \tag{23}$$

where a bar denotes a steady-state value and ζ is the elasticity of wages with respect to labor market conditions. The evolution of employment is procyclical and defined as

$$N_t = (1 - \delta)N_{t-1} + H_t \tag{24}$$

$$(1 - u_t) = (1 - \delta)(1 - u_{t-1}) + H_t \tag{25}$$

$$M_t = \frac{(1 - u_t) - (1 - \delta)(1 - u_{t-1})}{u_{t-1} + \delta(1 - u_{t-1})} \tag{26}$$

Market clearing and Equilibirum The demand for intermediary goods has to be equal to the total production of intermediary goods. Thus,

$$A_t \int_0^1 n_{J,t} dJ = \int_0^1 (\frac{p_{J,t}}{P_t})^{-\theta} dJ Y_t \tag{27}$$

I can ignore the integral on the right hand side of the equation since it is approximately equal to 1 after a first order approximation. Therefore, I obtain the aggregate production function

$$Y_t = A_t N_t \tag{28}$$

Labor market clearing requires

$$N_t = (1 - u_t) \tag{29}$$

where u_t is the unemployment rate and H_t is aggregate hiring. The profit of the intermediate producer is paid out in dividends to the households. Thus,

$$Y_t - \psi M_t H_t = (w_t + d_t)(1 - u_t) \tag{30}$$

The aggregate resource constraint has to hold

$$Y_t - \psi M_t H_t = \int c_t(a, e) d\Gamma_t(a, e) \tag{31}$$

Goods market clearing requires:

$$Y_t = \int c_t(a,e) d\Gamma_t(a,e) \tag{32}$$

The bond market clears:

$$\int g_t(a,e) d\Gamma_t(a,e) = B \tag{33}$$

The exogenous processes are defined as

$$logA_t = \rho_A logA_{t-1} + \epsilon_{A,t} \tag{34}$$

$$log\xi_t = \rho_\xi log\xi_{t-1} + \epsilon_{\xi,t} \tag{35}$$

Definition of an equilibrium An equilibrium of the model requires household policy rules for savings g_t and a cumulative distribution of wealth $\Gamma_t(a,e)$ that evolves according to the household's problem with the **actual** job finding M_t and separation rate δ. At each date, the vector of the aggregate variables $\{M_t, w_t, \tau_t, Y_t, u_t, d_t, i_t, R_t, \pi_t, p_t^A, p_t^B\}_{t=0}^{\infty}$ has to ensure in equilibrium that households maximize their utility and firms maximize their profits given their constraints, i.e. I obtain a value and policy function by solving the problem in (3), and (19) is fulfilled. Bond market and goods market clear, i.e. (31)

and (33) are fulfilled. Government budget constraint hold with equality, i.e. (20) is fulfilled.

5 Results

In this section, I present different specifications of my model with biased labor market expectations. In the first case, I use labor market expectations for the U.S. based on the calibration by Balleer et al. [2021], and in the second case, I incorporate the biases in the German labor market from Table 1. I solve the household's problem by iterating on the Euler equation backward through time using the endogenous gridpoint method of Carroll [2006] to compute the policy function and use Young [2010] method to solve for the stationary distribution. For each specification, I first solve the model in Python using the Reiter [2009] method to find the steady-state. One can find the stationary competitive equilibrium of this economy using the same techniques as in a Bewley-Huggett-Aiyagari type model where there is idiosyncratic uncertainty, but no aggregate shocks. To solve for the dynamics of the economy, I apply a first-order perturbation around the steady-state without aggregate shocks. The resulting solution is nonlinear with respect to the idiosyncratic variables, but linear with respect to the aggregate states.[5] Thus, the method cannot capture nonlinearities. Next, I solve for the transitory dynamics with time-iteration in response to an aggregate shock following Rendahl [2017]. For each specification, I present its steady-state wealth distribution as well as analyze the impulse responses of the main aggregate variables following a 1% transitory positive productivity shock and a 1% transitory contractionary monetary

[5]See Auclert et al. [2021] for a detailed explanatation for using Reiter's method for HANK models

policy shock.

5.1 Calibration

One period in the model corresponds to a quarter. I fix the steady-state real interest rate at 2 percent annually and adjust the discount factor to match this. I set the coefficient of risk aversion to 2. In my baseline calibration, I set the supply of government bonds, B, to match the ratio of aggregate liquid assets to GDP. For my choices of the desired markup of intermediate firms and probability of maintaining a fixed price, I follow Christiano et al. [2011] and set $\mu = 1.2$ and $\Theta = 0.15$ respectively. The implied degree of price stickiness is on the high side of values used in the business cycle literature. This tends to reduce the size of the effects I find on inflation since it makes inflation less sensitive to changes in current marginal costs. I set the other standard parameters to the values often used in the literature described in Table 2.[6] To calibrate the idiosyncratic process, I follow Balleer et al. [2021] and set the magnitude of job finding bias to 0.18 and the job separation bias to 0. In the second case I set the job finding bias to 0.1 and the job separation bias to 0.07 to match the characteristics of Germany. Regarding hiring costs, I follow Blatter et al. [2012] and set the hiring costs to 0.1. I set the matching efficiency, ζ, to 0.78 and 0.73 to get a realistic unemployment rate of 3% for the U.S. and 2% for Germany in the steady-state respectively. To capture heterogeneity between the job markets, I set the unemployment benefits to 30% and 60% of the wage in the steady-state for the U.S. and Germany respectively .

[6]Parameters for the second case in parentheses

Parameter:	Description:	Value:	Source:
β	Discount factor	0.97 (0.99)	
γ	Risk aversion	2	
B	Supply of bonds	1.4 x annual GDP	
ζ	Matching efficiency	0.78 (0.73)	
ψ	Hiring costs	0.1	Blatter et al. [2012]
μ	Mark up	1.2	Christiano et al. [2011]
Θ	Firm price adjustment	0.15	Christiano et al. [2011]
δ	Job separation rate	0.15	Balleer et al. [2021]
θ	Elasticity of Substitution	6	
Δ_1	Job separation bias	0 (0.07)	Balleer et al. [2021]
Δ_2	Job finding bias	0.18 (0.1)	Balleer et al. [2021]
b	Unemployment rate	0.3 x $\bar{w}$ (0.6 x $\bar{w}$)	
ρ_A	Persistence of productivity shock	0.9	
ρ_ξ	Persistence of monetary policy shock	0.8	

Table 2: Parameters

Steady-state wealth distribution For the first case, where I set $\Delta_2 = 0.18$ and $\Delta_1 = 0$, Figure 3 describes the steady-state wealth distribution, and Table 3 illustrates the steady-state values for the most important aggregates.[7] Since the lower expected frequency of adverse labor market shocks weakens the motive to save out of precautionary reasons, over-optimistic agents hold fewer assets than their rational counterpart, and an optimistic bias in expectations leads to higher wealth inequality. As Figure 3 suggests, the Gini coefficient increased by 2.1% compared to the model with rational expectations, indicating that biased labor market expectations contribute to greater inequality in the economy. This rise in inequality is further illustrated by the fact that the mass of agents sitting at the borrowing constraint increased by 247.7%. The mass of HtM agents has increased by 251.2% and the average MPC has increased by 16.9% compared to the model with rational expectations.

For the second case, where I set $\Delta_2 = 0.1$ and $\Delta_1 = 0.07$, Figure 5 describes the steady-state wealth distribution, and Table 3 illustrates the steady-state values for the most important aggregates. The Gini-coefficient decreased by 5.1% compared to the model

[7] steady-state values for the second case are in parentheses.

with rational expectations. Thus, under this calibration, inequality decreased compared to the first case as Figure 4 illustrates. The mass of agents sitting at the borrowing constraint decreased by 90.1% compared to the model with rational expectation. The mass of HtM agents has decreased by 30.6% and the average MPC has decreased by 8.7% compared to the model with rational expectations.

The steady-state wealth distributions for the two cases are different since most of the mass of agents in the model is in the employment state. Thus, the over-pessimism to lose the current occupation, which raises the precautionary savings motive, outweighs the effects of optimism to find a job when unemployed.

Variables:	Notation:	steady-state value:
Output	Y	0.94 (0.91)
Consumption	C	0.94 (0.91)
TFP	Z	1 (1)
Gross inflation	Π	1 (1)
Job finding rate	M	0.64 (0.53)
Unemployment rate	M	0.03 (0.02)
Real interest rate	r	1.005 (1.005)
Nominal interest rate	i	1.005 (1.005)

Table 3: Values of the main aggregate variables in steady-state

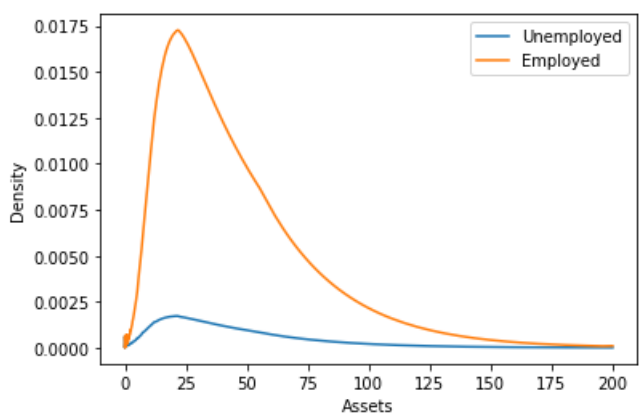

Figure 2: First case: steady-state conditional wealth distribution with rational expectation

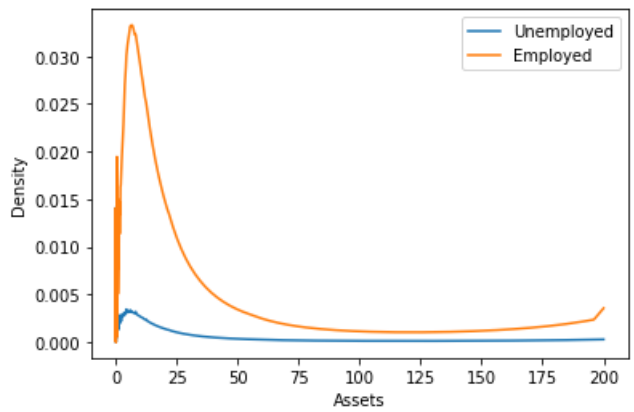

Figure 3: First case: steady-state conditional wealth distribution with biased labor market expectation

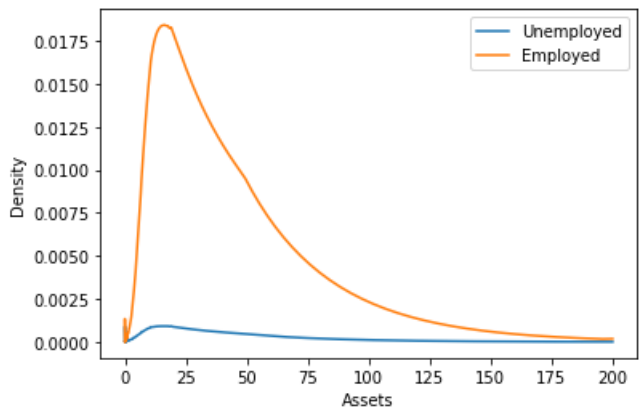

Figure 4: Second case: steady-state conditional wealth distribution with rational expectation

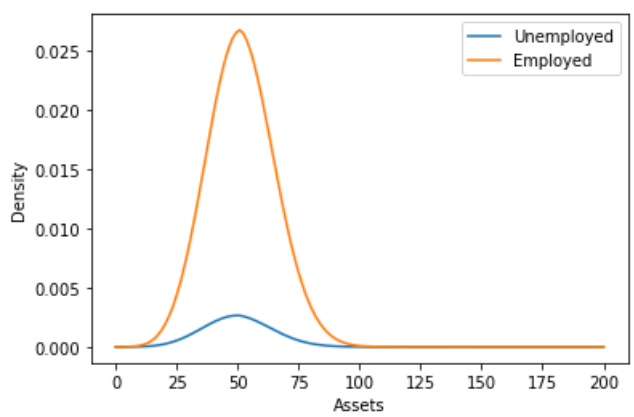

Figure 5: Second case: steady-state conditional wealth distribution with biased labor market expectation

5.2 Technology Shock

I assume that the economy is in steady-state and I implement a 1% transitory positive technology shock, which implies that $\epsilon_{A,0}$ changes from 0 to 1, and therefore, A_0 increases by 1% at the time of the shock. Figure 6 display the corresponding impulse responses for the main aggregate variables. Table 4 presents the moments of the simulated model for these variables. Following a 1% transitory positive technology shock, A_0 increases, which increases output immediately, increases wages, decreases firm's marginal costs and thus prices, i.e. the inflation rate. As a consequence, the central bank decreases the nominal rate i_t. Through the increase in productivity, firms hire more workers and this results in an increase in the endogenous job finding rate, M_t, and therefore unemployment rate, u_t, decreases.

In the first case, compared to the case without biased labor market expectations, there is more mass is at the borrowing constraint in the steady-state. This implies that there is a higher proportion of households with high MPCs compared to the case without biased labor market expectations. The differences in the impulse response functions due to a technology shocks are driven by the two factors: (I) different steady-state distributions, as explained above; (II) the upward shift of the consumption function, as Figure 1 suggests. These two effects lead to an amplification of the responses of the unemployment rate, endogenous job finding rate; and dampens the responses of the real and nominal rate, and inflation rate.

In the second case, Figure 7 reveals reversed effects. Contrary to the first case, there is a substantial decrease in the mass of agents constrained by the borrowing limit. The presence of a positive job separation and job finding bias, as indicated in Figure 1,

results in a downward shift of the consumption function. This occurs because more agents are in the employment state, and therefore the impact of the job separation bias outweighs that of the job finding bias on the consumption behavior, and therefore a positive technology shock leads to a smaller increase in aggregate demand than in the model with rational expectation. This explains the reversed effects compared to the first case. Table 5 presents the moments of the simulated model.

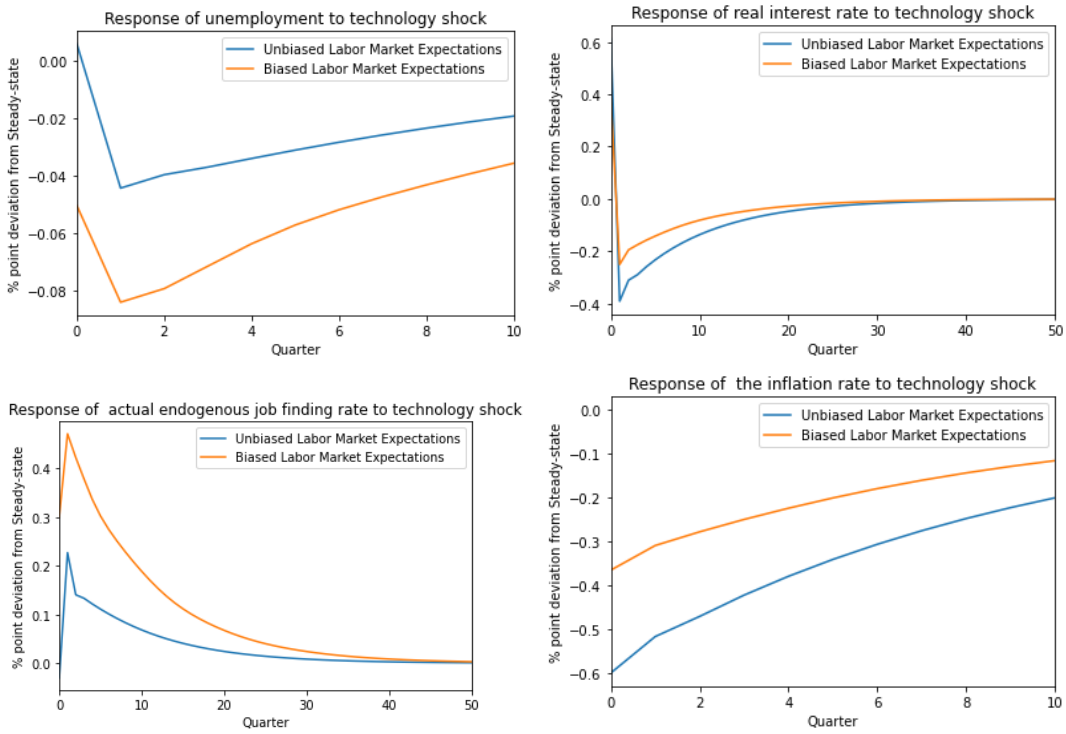

Figure 6: First case: 1 % positive transitory technology shock

Variable:	Mean:	Standard deviation:
Y	0.024	0.012
u	-0.067	0.012
R	-0.039	0.016
i	0.003	0.056
Π	-0.088	0.047
M	0.0834	0.021

Table 4: First case: Moments of simulated model following a 1 % positive transitory technology shock

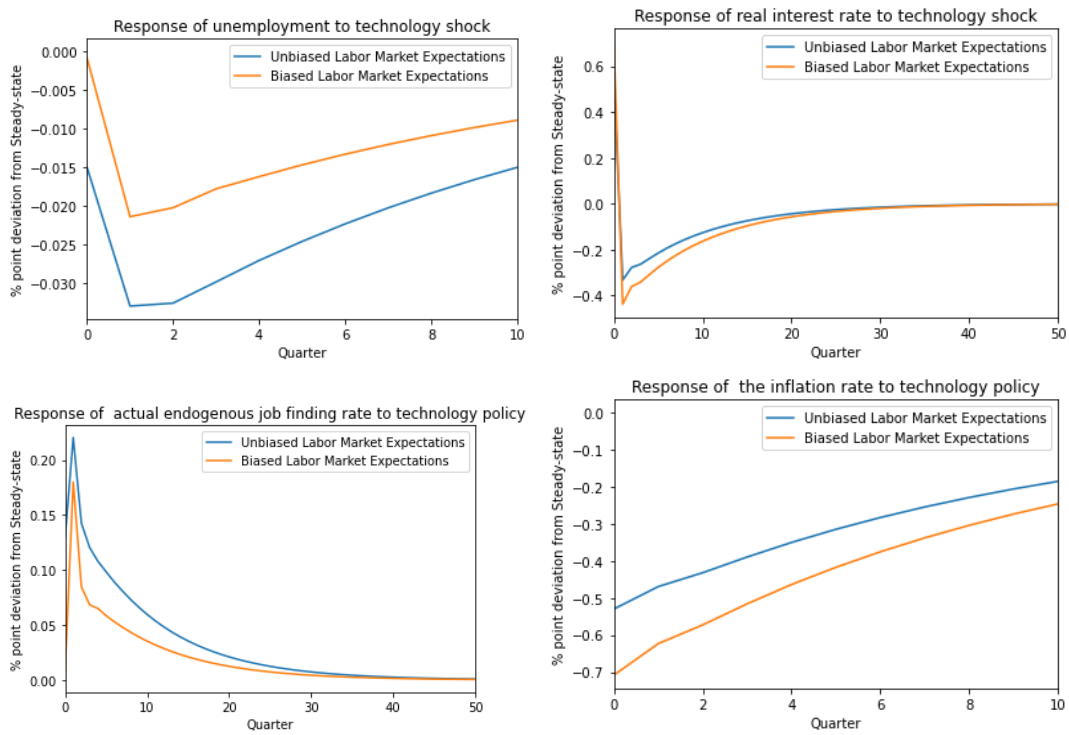

Figure 7: Second case: 1 % positive transitory technology shock

Variable:	Mean:	Standard deviation:
Y	0.004	0.013
u	-0.005	0.013
R	-0.025	0.018
i	0.072	0.029
Π	-0.045	0.039
M	0.023	0.005

Table 5: Second case: Moments of simulated model following a 1 % positive transitory technology shock

5.3 Monetary Policy

I again assume that the economy is in steady-state and I implement a 1% contractionnary monetary policy shock, so the monetary policy shock $\epsilon_{\xi,0}$ changes from 0 to 1. Following a 1% contractionnary monetary policy shock, $\xi_{\epsilon,0}$ enters the model through the Taylor rule which leads to an increase in the nominal rate, and therefore the real interest rate increases. Due to this, the income tax, τ_t, increases to fulfill govern-

ment's budget balance. Because of the rise in the real interest rate and income tax, consumption decreases. From the Euler-equation, this occurs because it becomes more advantageous to defer consumption today in order to consume more in the future due to the higher real interest rate. As a response of the decrease in aggregate demand, firms produce less, and consequently, there is a reduction in hiring, leading to a decrease in the endogenous job-finding rate. As a result, inflation decreases, wages decrease, and dividends decrease.

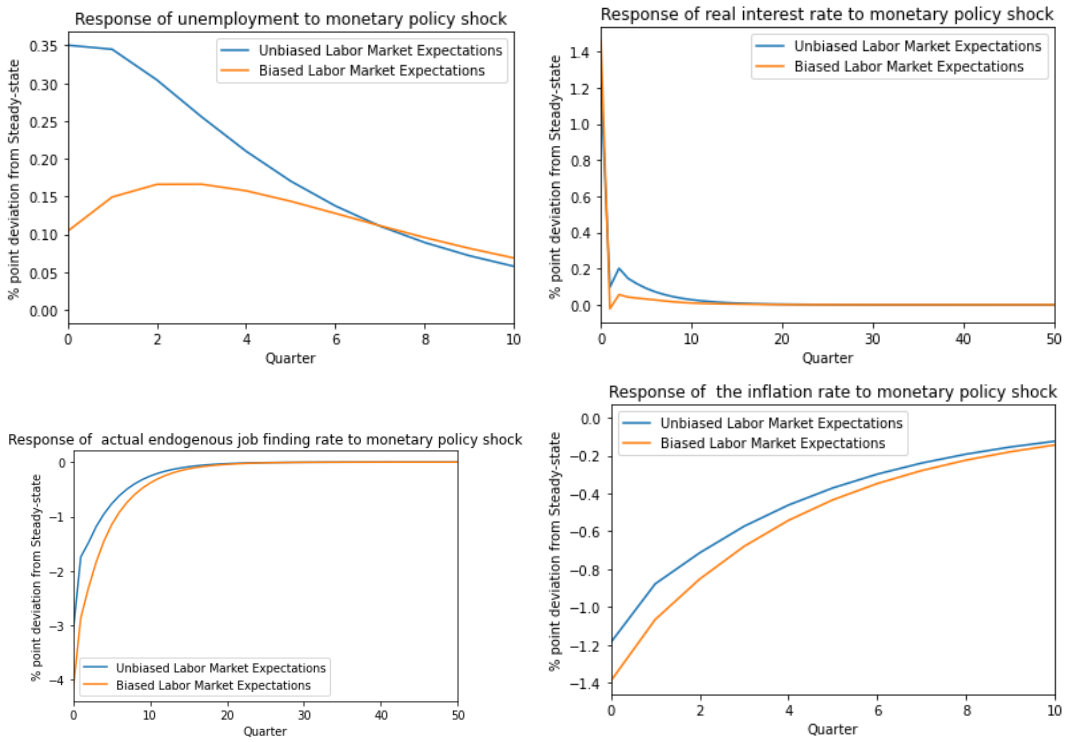

Figure 8: First case: 1 % transitory monetary policy shock

In the first case, Figure 8 displays the corresponding impulse responses for the main aggregate variables. Table 6 presents the moments of the simulated model for these variables. The monetary policy shock affects the households through the budget constraint. The agent receives a bigger value of assets than expected if she has accumulated a positive stock of savings at the time of the shock. However, biased labor market ex-

pectations lead to smaller savings. Consequently, there are more people with no asset holding, i.e. $a = 0$, who are not affected by the monetary policy shock initially through the savings channel, and the overall savings level is smaller in the aggregate. The monetary policy shock increases the nominal rate, in turn the real rate, and increases the income tax of the same magnitude as in the model with rational expectation. Household consumption decreases by a smaller amount than in the rational expectation case which is due to the asset holding described above, which is in line with Farhi and Werning [2019] and Pfäuti et al. [2023]. However, choosing price stickiness in HANK makes profits countercyclical, which results in an amplification of the responses for the nominal rate, inflation rate, and endogenous job finding rate.[8]

The second case exhibits reversed effects in Figure 9. The increased precautionary savings boosts aggregate savings. The increase in the real rate leads to an higher absolute increase of the asset value. Through that, there is an larger increase of savings in absolute terms which decreases aggregate consumption more than in the rational expectation model.

Variable:	Mean:	Standard deviation:
Y	-0.022	0.003
u	0.022	0.003
R	0.0188	0.0004
i	-0.0047	0.002
Π	-0.065	0.001
M	-0.17	0.005

Table 6: First case: Moments of simulated model following a 1 % transitory monetary policy shock

[8]The dynamics of the inflation rate, unemployment, nominal and real interest rate are not in line with Farhi and Werning [2019] and Pfäuti et al. [2023].

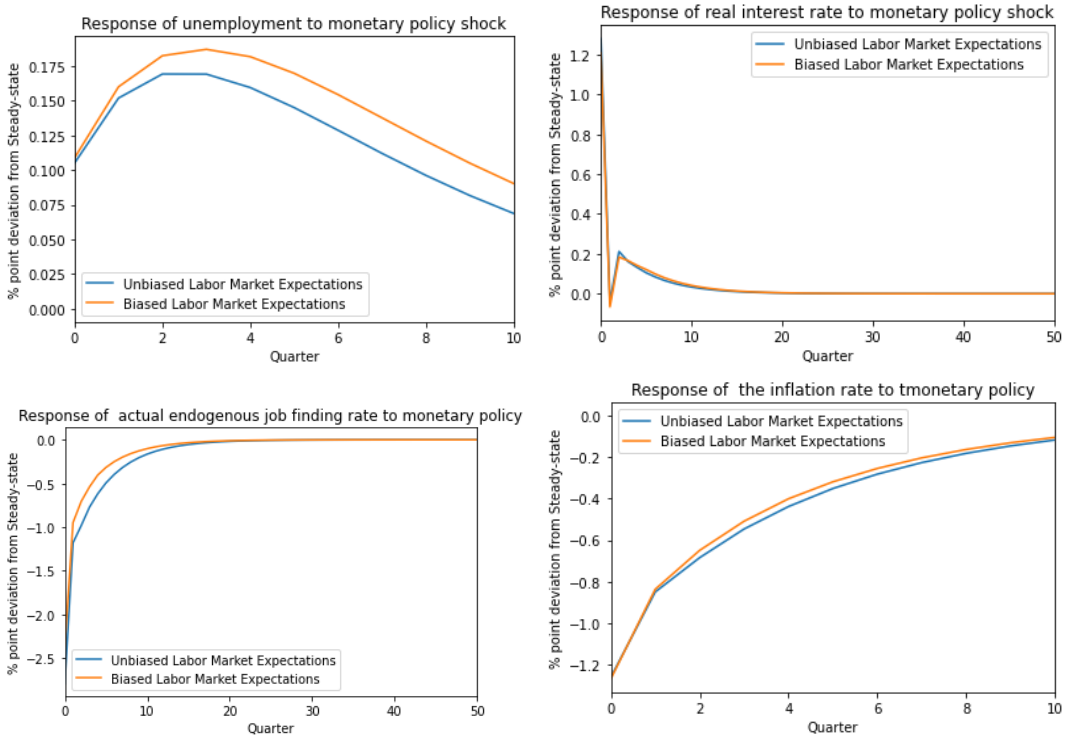

Figure 9: Second case: 1 % transitory monetary policy shock

Variable:	Mean:	Standard deviation:
Y	-0.047	0.004
u	0.047	0.004
R	0.028	0.013
i	-0.044	0.04
Π	-0.062	0.02
M	-0.261	0.001

Table 7: Second case: Moments of simulated model following a 1 % transitory monetary policy shock

6 Conclusion

This paper incorporates biased labor market expectations into a HANK framework. By studying two types of shocks — a positive technology shock and a contractionary monetary policy shock — I demonstrate how biased labor market expectations affect the impulse response functions of the main aggregate variables in the model using

two different calibrations. The two calibrations exhibit reverse effects when simulating aggregate shocks. These results contribute to a growing literature that incorporates non-rational expectations in a HANK framework to explain stylized facts in the data. Furthermore, this paper also analyzes a rather recent survey data set collected by the ECB. I show that all six countries' survey participants exhibit biased labor market expectations. However, due to the limited number of available waves, the results should be taken with a grain of salt.

Outlook for future research At this point, I consider studying the "forward guidance puzzle" with biased labor market expectations as interesting since monetary policy is already less effective in the first case. Further, Auclert and Rognlie [2018] show that higher inequality leads to smaller output, and thus it might be interesting to study welfare losses over the business cycle due to biased expectations since the Gini-coefficient increased in both cases compared to the model with rational expectation.

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7 Appendix

Table 8: Expected job finding rate conditional on household's characteristics

	Country	Gender	Level of Educ.	Exp. job find. rate given searching	variance	n
1	BE	1	1	19.30	532.01	40
2	BE	1	2	22.07	707.23	44
3	BE	1	3	33.12	1292.14	67
4	BE	2	1	13.25	243.57	20
5	BE	2	2	22.48	623.17	58
6	BE	2	3	38.74	1163.93	68
7	DE	1	1	42.69	1049.79	72
8	DE	1	2	28.66	930.52	109
9	DE	1	3	28.14	682.75	143
10	DE	2	1	23.77	433.75	47
11	DE	2	2	26.28	899.02	79
12	DE	2	3	35.46	910.45	81
13	ES	1	1	34.43	1054.95	164
14	ES	1	2	34.88	906.96	146
15	ES	1	3	33.13	918.20	237
16	ES	2	1	34.24	828.21	274
17	ES	2	2	28.27	942.74	199
18	ES	2	3	35.17	915.56	523
19	FR	1	1	35.91	959.99	45
20	FR	1	2	28.67	970.77	93
21	FR	1	3	35.46	1057.80	226
22	FR	2	1	38.24	1241.97	37
23	FR	2	2	24.44	586.22	156
24	FR	2	3	35.48	921.44	266
25	IT	1	1	29.97	701.33	70
26	IT	1	2	27.50	924.54	304
27	IT	1	3	29.98	701.30	174
28	IT	2	1	17.74	545.98	97
29	IT	2	2	19.54	613.93	482
30	IT	2	3	27.29	814.02	296
31	NL	1	1	50.86	1447.48	7
32	NL	1	2	25.89	739.30	36
33	NL	1	3	23.85	812.43	41
34	NL	2	1	10.83	413.33	23
35	NL	2	2	21.47	750.74	30
36	NL	2	3	32.97	618.76	34

Table 9: Expected job finding rate conditional on household's characteristics and searching

	Country	Gender	Level of Educ.	Exp. job find. given not searching	variance	n
1	BE	1	1	15.75	198.92	4
2	BE	1	2	17.55	383.88	22
3	BE	1	3	16.08	619.91	24
4	BE	2	1	13.37	724.58	19
5	BE	2	2	9.20	233.54	20
6	BE	2	3	14.76	410.62	42
7	DE	1	1	10.33	358.23	24
8	DE	1	2	17.41	446.69	69
9	DE	1	3	23.97	664.24	39
10	DE	2	1	11.04	325.12	25
11	DE	2	2	10.86	332.08	42
12	DE	2	3	12.95	384.50	41
13	ES	1	1	35.06	998.14	51
14	ES	1	2	19.94	660.13	31
15	ES	1	3	24.64	851.95	64
16	ES	2	1	18.71	447.99	112
17	ES	2	2	23.31	560.71	74
18	ES	2	3	26.79	768.60	175
19	FR	1	1	12.90	391.24	29
20	FR	1	2	24.90	570.30	39
21	FR	1	3	19.05	513.42	81
22	FR	2	1	5.70	68.84	30
23	FR	2	2	14.95	524.62	85
24	FR	2	3	15.28	394.58	150
25	IT	1	1	19.88	631.95	26
26	IT	1	2	20.99	777.97	105
27	IT	1	3	20.92	657.75	50
28	IT	2	1	15.81	539.80	86
29	IT	2	2	13.69	408.50	261
30	IT	2	3	20.82	681.26	152
31	NL	1	1	12.75	200.25	4
32	NL	1	2	16.15	657.71	20
33	NL	1	3	3.84	23.47	19
34	NL	2	1	14.70	480.96	20
35	NL	2	2	18.27	332.82	11
36	NL	2	3	15.35	274.12	17

Table 10: Expected job finding rate conditional on household's characteristics

	Country	Gender	Level of Educ.	Exp. job find. aggregate	variance	n
1	BE	1	1	18.58	493.34	45
2	BE	1	2	21.11	644.45	70
3	BE	1	3	29.55	1218.15	98
4	BE	2	1	13.02	443.67	41
5	BE	2	2	18.51	541.60	81
6	BE	2	3	31.19	1066.96	127
7	DE	1	1	32.75	1004.11	107
8	DE	1	2	24.95	787.43	189
9	DE	1	3	28.40	679.05	198
10	DE	2	1	21.75	497.32	84
11	DE	2	2	21.02	720.54	129
12	DE	2	3	27.99	841.24	129
13	ES	1	1	36.32	1085.80	237
14	ES	1	2	32.19	885.86	188
15	ES	1	3	32.02	946.31	333
16	ES	2	1	30.00	758.47	413
17	ES	2	2	27.27	840.72	287
18	ES	2	3	33.14	902.56	748
19	FR	1	1	27.53	833.96	77
20	FR	1	2	29.44	906.15	142
21	FR	1	3	32.18	978.01	334
22	FR	2	1	23.85	965.09	74
23	FR	2	2	21.74	611.41	258
24	FR	2	3	28.95	846.74	440
25	IT	1	1	27.01	693.84	97
26	IT	1	2	26.29	918.49	429
27	IT	1	3	27.92	704.86	237
28	IT	2	1	16.41	522.65	196
29	IT	2	2	17.70	564.32	794
30	IT	2	3	25.36	782.86	479
31	NL	1	1	45.15	1494.64	13
32	NL	1	2	21.91	664.82	64
33	NL	1	3	18.34	734.14	68
34	NL	2	1	12.40	419.43	45
35	NL	2	2	18.78	607.49	45
36	NL	2	3	24.84	551.81	57

Table 11: Expected job separation rate conditional on household's characteristics

	Country	Gender	Level of Educ.	Exp. job separation rate	variance	n
1	BE	1	1	16.09	683.86	145
2	BE	1	2	8.44	298.09	824
3	BE	1	3	9.01	314.81	1690
4	BE	2	1	8.92	272.79	186
5	BE	2	2	8.84	374.98	794
6	BE	2	3	8.63	311.42	1713
7	DE	1	1	11.64	415.57	646
8	DE	1	2	8.23	296.91	2449
9	DE	1	3	6.87	200.99	4913
10	DE	2	1	8.74	261.31	651
11	DE	2	2	8.59	310.62	2650
12	DE	2	3	8.28	282.27	3288
13	ES	1	1	16.82	606.05	1661
14	ES	1	2	15.13	524.34	1326
15	ES	1	3	14.00	454.58	5560
16	ES	2	1	18.88	636.01	1450
17	ES	2	2	19.20	727.98	1113
18	ES	2	3	16.96	613.31	5112
19	FR	1	1	8.27	225.10	549
20	FR	1	2	8.53	277.87	1420
21	FR	1	3	8.57	260.16	5509
22	FR	2	1	9.23	311.42	380
23	FR	2	2	11.12	448.70	1533
24	FR	2	3	9.03	333.00	5338
25	IT	1	1	12.99	562.31	437
26	IT	1	2	10.15	382.21	3728
27	IT	1	3	9.19	341.86	4258
28	IT	2	1	13.74	575.68	396
29	IT	2	2	14.05	604.79	3151
30	IT	2	3	11.68	482.34	4011
31	NL	1	1	8.58	407.10	264
32	NL	1	2	6.79	280.41	964
33	NL	1	3	7.76	281.54	1564
34	NL	2	1	9.83	567.93	278
35	NL	2	2	8.68	412.11	976
36	NL	2	3	7.61	298.76	1366

Table 12: Regression 1

	(1)
(Intercept)	-54.65*** (std.error)
quintile2	-0.63 (std.error)
quintile3	-3.23 (std.error)
quintile4	1.75 (std.error)
quintileq5	12.22*** (std.error)
Education2	14.99*** (std.error)
Education3	16.64*** (std.error)
Female	-1.18 (std.error)
GER	0.95 (std.error)
ES	19.17*** (std.error)
FR	-8.73** (std.error)
IT	21.44*** (std.error)
NL	15.13*** (std.error)
Partner=1	-3.56* (std.error)
Num.Obs.	75503
R2	0.005
R2 Adj.	0.004
AIC	1017526.8
BIC	1017665.3
RMSE	204.22

Table 13: Regression 2

	(1)
(Intercept)	26.20*** (std.error)
quintile2	-1.28 (std.error)
quintile3	1.15 (std.error)
quintile4	0.31 (std.error)
quintile5	4.72** (std.error)
Education2	-1.88* (std.error)
Education3	2.96** (std.error)
Female	-3.48*** (std.error)
GER	1.99 (std.error)
ES	8.73*** (std.error)
FR	3.91* (std.error)
IT	0.17 (std.error)
NL	-3.03 (std.error)
Partner=1	-3.37*** (std.error)
Num.Obs.	7353
R2	0.035
R2 Adj.	0.034
AIC	70001.5
BIC	70105.0
RMSE	28.19

“I have written the present thesis myself and have used exclusively the sources and aides mentioned. This thesis has not yet been submitted as an examination in another degree programme. All passages taken word-by-word or the meaning of which are quoted from published or unpublished texts, as well as all indications based on oral accounts, have been marked as such.”

Frankfurt, 24.10.2023